

RetroDiff: Training Objective Derivation

We derive the training objective for RetroDiff by starting from the standard variational lower bound on the log-likelihood of a diffusion model. Let x denote a clean data sample, x_t its noised version at diffusion step $t \in \{1, \dots, T\}$, q the fixed forward noising process, and p_θ the learned reverse process with parameters θ .

Step 1 — Variational lower bound (ELBO). Applying Jensen’s inequality to the intractable log-marginal $\log p_\theta(x)$ yields the evidence lower bound:

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)} \right] = -\mathcal{L}_{\text{ELBO}} \quad (1)$$

Maximising the right-hand side is equivalent to minimising $\mathcal{L}_{\text{ELBO}}$, which we expand next.

Step 2 — Reconstruction + KL decomposition. Factoring the joint distributions and applying the Markov structure of the forward process decomposes (1) into a reconstruction term and a sum of per-step KL divergences:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0 | x_1)]}_{\text{reconstruction}} + \sum_{t=2}^T \underbrace{\text{KL}(q(x_{t-1} | x_t, x) \parallel p_\theta(x_{t-1} | x_t))}_{\text{step-}t \text{ denoising}} \quad (2)$$

Each KL term measures how well the reverse step $p_\theta(x_{t-1}|x_t)$ matches the tractable posterior $q(x_{t-1}|x_t, x)$; the reconstruction term handles the final generation step.

Step 3 — Retrieval-augmented objective (RetroDiff). RetroDiff’s key contribution is to condition the entire ELBO on a set $R(x)$ of k exemplars retrieved from the training corpus based on their similarity to x . The per-sample objective is then averaged over the data distribution:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}(x \mid R(x))] \quad (3)$$

Equation (3) reduces to the standard diffusion objective when $R(x) = \emptyset$, making RetroDiff a strict generalisation of the baseline. In practice $R(x)$ is computed once per sample with a frozen retriever and cached for the duration of training.